DIMENSIONAL ANALYSIS ON ONE PAGE

The density of the universe from units alone

Gaussian elimination on a matrix of exponents delivers $\rho \sim H^2/G$, within a decade of the Friedmann critical density.

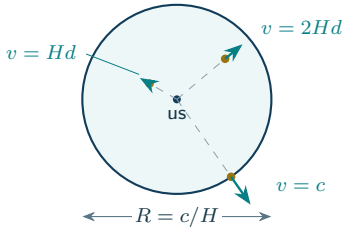
1 / THE RULE AND THE PICTURE

Every quantity is a vector of exponents of $\mathrm{M}, \mathrm{L}, \mathrm{T}$

A product $\rho^a H^b G^g c^k h^e$ is free of units exactly when the exponents of mass, length and time all cancel, which is the linear system

$$A\mathbf{x} = \mathbf{0}, \qquad \mathbf{x} = (a, b, g, k, e)^{\mathsf{T}}$$

Each column of A lists the $(\mathrm{M}, \mathrm{L}, \mathrm{T})$ exponents of one quantity, read from its own defining equation, and n quantities with $\operatorname{rank} A = r$ give $n - r$ dimensionless groups.



quantity	defining equation	rearranged	$(\mathrm{M}, \mathrm{L}, \mathrm{T})$
ρ density	$\rho = m/V$	$\mathrm{M/L^3}$	$(1, -3, 0)$
H Hubble	$v = Hd$	$(\mathrm{LT^{-1}})/\mathrm{L} = \mathrm{T^{-1}}$	$(0, 0, -1)$
G Newton	$F = Gm_1m_2/r^2$	$(\mathrm{MLT^{-2}})\mathrm{L^2/M^2}$	$(-1, 3, -2)$
c light	$c = d/t$	$\mathrm{L/T}$	$(0, 1, -1)$
h Planck	$E = h\nu$	$(\mathrm{ML^2T^{-2}})/\mathrm{T^{-1}}$	$(1, 2, -1)$

Force is $\mathrm{MLT^{-2}}$ and energy $\mathrm{ML^2T^{-2}}$ from $F = ma$ and $E = Fd$; $\hbar = h/2\pi$ changes no exponent. The picture shows H as a speed per distance, reaching c at $R = c/H$.

2 / ELIMINATE WITH h INCLUDED

Five quantities, rank 3, two groups

Columns ρ, H, G, c, h and rows $\mathrm{M}, \mathrm{L}, \mathrm{T}$. Add $3\times$ the M row to the L row, then swap the L and T rows and negate the new second row:

	ρ	H	G	c	h
M	1	0	−1	0	1
L	−3	0	3	1	2
T	0	−1	−2	−1	−1

$\Longrightarrow$

	ρ	H	G	c	h
M	**1**	0	−1	0	1
L	0	**1**	2	1	1
T	0	0	0	**1**	5

The pivots (teal) sit in ρ, H, c, leaving G and h free (gold). Back-substitution from the bottom gives $k = -5e$, then $b = -2g + 4e$, then $a = g - e$, so that

$$\mathbf{x} = \mathbf{g}\,(1, -2, 1, 0, 0)^{\mathsf{T}} + \mathbf{e}\,(-1, 4, 0, -5, 1)^{\mathsf{T}}.$$

$$\Pi_A = \tfrac{G\rho}{H^2}, \quad \Pi_B = \tfrac{hH^4}{\rho c^5}, \quad \Pi_A\Pi_B = \tfrac{hGH^2}{c^5}$$
$$\rho = \frac{H^2}{G}\, F\!\left(\tfrac{hGH^2}{c^5}\right)$$

3 / CLOSE THE SYSTEM: LEAVE h OUT

Four quantities, rank 3, one group

Without Planck's constant the matrix loses its last column, and the same first step, $\mathrm{L} \leftarrow \mathrm{L} + 3\mathrm{M}$, leaves the row $(0, 0, 0, 1)$, which reads $k = 0$:

	ρ	H	G	c
M	1	0	−1	0
L	−3	0	3	1
T	0	−1	−2	−1

$\Longrightarrow$

	ρ	H	G	c
M	**1**	0	−1	0
L	0	**1**	2	1
T	0	0	0	**1**

The speed of light is eliminated by its own row, because length can only be balanced between ρ and G. Only G is free, so $\mathbf{x} = \mathbf{g}\,(1, -2, 1, 0)^{\mathsf{T}}$ and the single group is $\Pi = G\rho/H^2$.

$$\rho = \Pi\,\frac{H^2}{G}$$

with Π a pure number that units cannot fix

Check: $[H^2/G] = \mathrm{T^{-2}} \cdot \mathrm{ML^{-3}T^2} = [\rho]$.

4 / PUT IN A NUMBER

What units alone predict

With $H_0 \approx 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}} = 2.27 \times 10^{-18}\ \mathrm{s^{-1}}$ and $G = 6.67 \times 10^{-11}$ in SI units,

$$\frac{H^2}{G} \approx 7.7 \times 10^{-26}\ \mathrm{kg\,m^{-3}} \approx 46 \text{ protons per cubic metre.}$$

Only H and G were used, yet the answer lands in the right decade.

5 / THE FRIEDMANN VALUE OF Π

The pure number from full general relativity

For a spatially flat universe $H^2 = \frac{8\pi G}{3}\rho$, so $\rho_c = \dfrac{3}{8\pi}\dfrac{H^2}{G}$ and $\Pi = \dfrac{3}{8\pi} \approx 0.119$, giving $\rho_c \approx 9.2 \times 10^{-27}\ \mathrm{kg\,m^{-3}}$, about 5.5 protons per cubic metre. The dimensional estimate therefore overshoots by $8\pi/3 \approx 8.4$, less than one decade.

ASIDE / THE HUBBLE SPHERE

Ask that the mass inside $R = c/H$ just reaches its Schwarzschild limit, $2GM/(Rc^2) = 1$. Then $M = c^3/(2GH)$ and $\rho = \dfrac{M}{\frac{4}{3}\pi R^3} = \dfrac{3H^2}{8\pi G}$ exactly, with the powers of c cancelling just as the elimination predicted.

ASIDE / WHY h STAYS OUT

The group $hGH^2/c^5 = (H\ell_P/c)^2$ is of order 10^{-122}, the ratio of H^2/G to the Planck density $c^5/(\hbar G^2)$. Friedmann's equation contains no h, so F is a constant; a quantum estimate with $F \sim \Pi^{-1}$ would instead return the Planck density, the familiar vacuum-energy discrepancy of about 122 orders of magnitude.

Convention. Buckingham's Π theorem: n quantities in r independent dimensions yield $n - r$ groups, the null space of A. In Python, `sympy.Matrix(A).nullspace()` returns the two vectors of panel 2. **Companions:** sqrt-time rule with dice; the Kepler log–log sheet; *How Many Scales Can You Build?*

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